









WHAT THIS SHEET IS FOR, AND WHAT IT LEAVES OPEN

This page is the only place the ten sheets are joined. Each sheet symbol carries one hierarchical pin per net its child exports, and each pin carries a stub with a label: two labels anywhere on this page with the same name are one net. hw/check_netlist.py exports the whole hierarchy and checks it, the way sim/run_sim.py checks the values.

THE PINOUT IS BOSCH'S (8 Nov 2026, refs/bosch-pinout-md1cs162.md, decisions log 18). No permanent battery feed: pin 71 (ignition) closes the main relay through a wake FET, the relay's contact feeds pins 04/06 -- the board's only supply -- and the MCU holds pin 50 through the after-run. Pins 01/02 are the power ground. Pin 21 is the water-in-fuel input, 09 the GCU sync input, 14 diagnosis request, 25 the stop switch.

No net reaches exactly one pin: every single-pin net that was recorded here has closed, the last two (the power ground and the trip loop) with the Bosch pinout. What is still open lives off this page: the connector's land pattern and the housing (field brief Group 6), and the trip module's wiring to the main relay's coil feed.

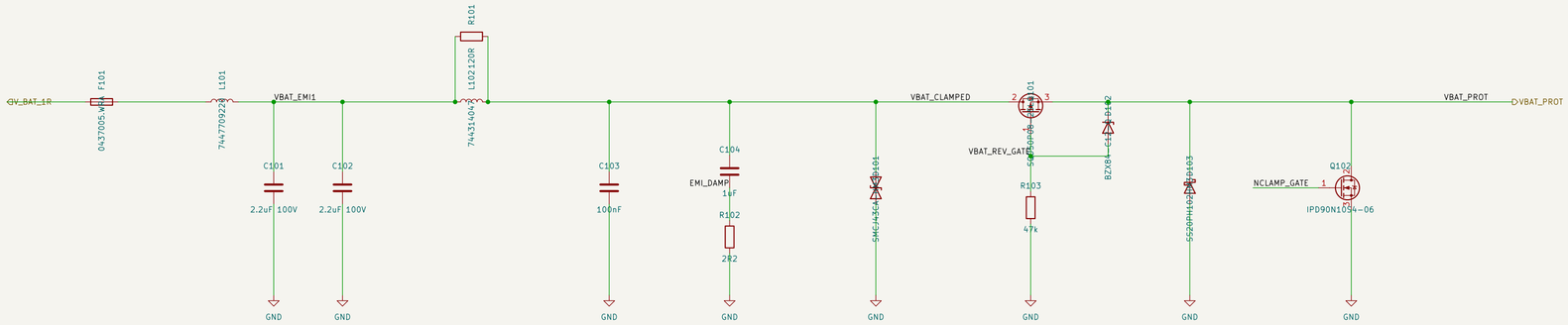
Every sheet maps to blocks in sim/blocks/ and tags in docs/bom_requirements.md
Root sheet -- generated by hw/gen_project.py

ecu25kva

Sheet: /
File: ecu25kva.kicad_sch

Title: Genset ECU -- 25 kVA Kirloskar 3GK550ETA 4SR1

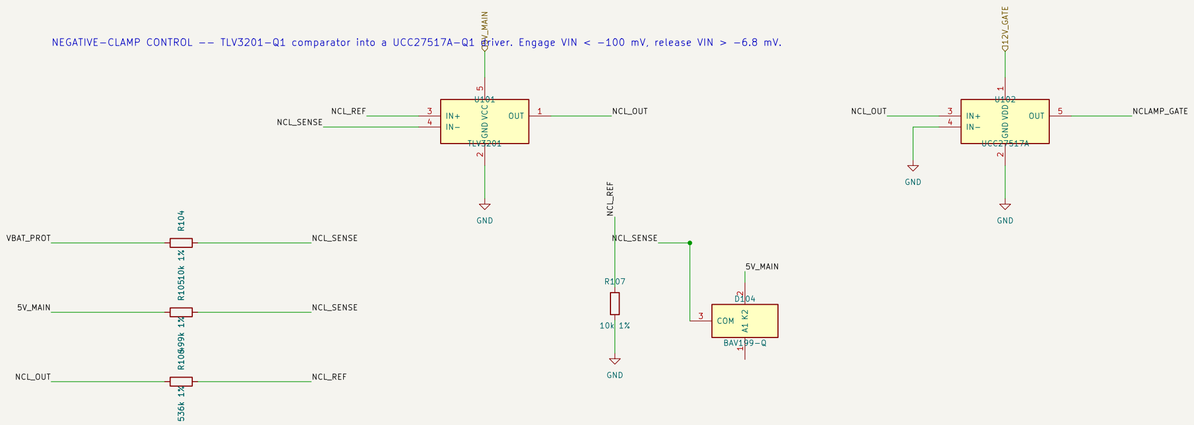
Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 1/11



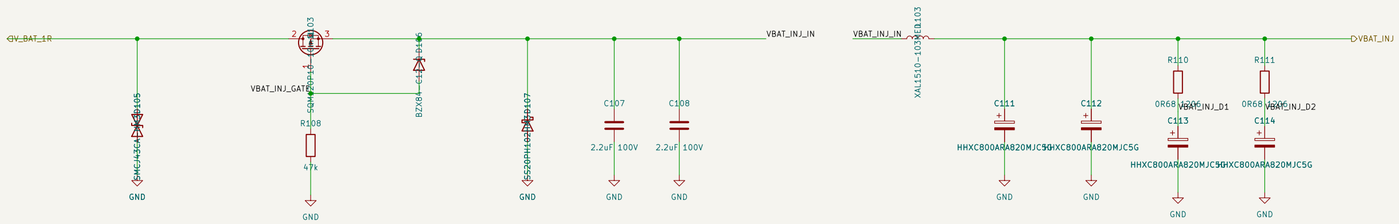
GATE DRIVE -- both power-path gates are driven:\n VBAT_REV_GATE Q1 is a P-FET: 47k gate to ground, zener gate-to-source. No controller needed.\n NCLAMP_GATE Q2 (negative clamp): TLV3201-Q1 comparator into a UCC27517A-Q1 driver.\n Engage below -100 mV, release above -5 mV (see neg_clamp_control).\nKeep U102 at Q2's gate with a short return to Q2's source: the clamp's speed is the gate loop's.

Stage order is spec sec.4: EMI filter -> fuse -> TVS -> reverse-battery FET -> buck VIN.\nEMI filter values are emi_filter.cir element for element. Every part carries its source in a hidden Source field.

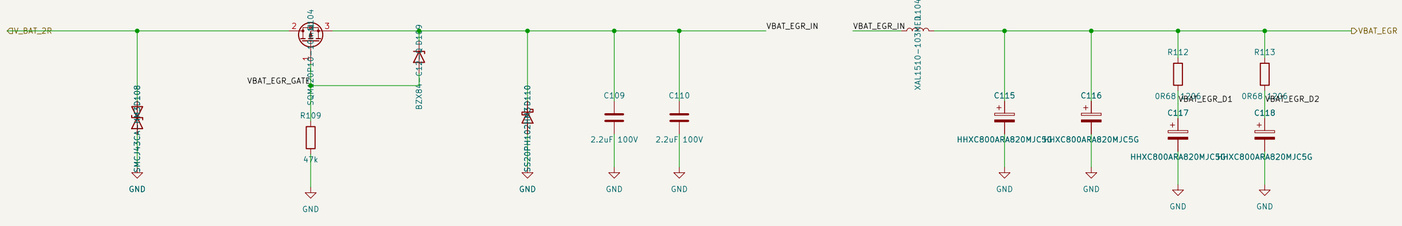
NEGATIVE-CLAMP CONTROL -- TLV3201-Q1 comparator into a UCC27517A-Q1 driver. Engage VIN < -100 mV, release VIN > -6.8 mV.



PIN 04 -> VBAT_INJ: injector hold switches and the boost converter input. 10 A loom fuse.



PIN 06 -> VBAT_EGR: the EGR H-bridge. 10 A loom fuse.



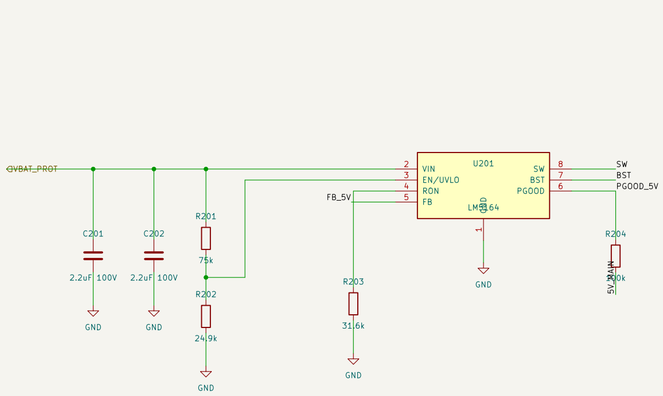
Q1: P-FET, passive gate. Q2: comparator + driver, a low-side ideal diode that MUST release
TVS and clamp: docs/bom_requirements.md tags TVS, NEG-CLAMP
EMI filter values are emi_filter.cir element for element
Generated by hw/gen_sheet_power_input.py -- do not hand-edit until it is retired

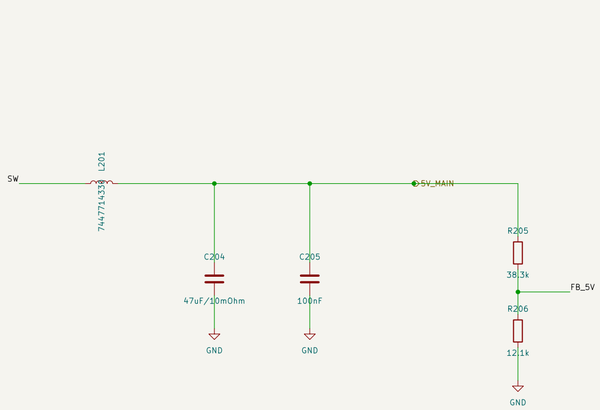
ecu25kva

Sheet: /power_input/
File: power_input.kicad_sch

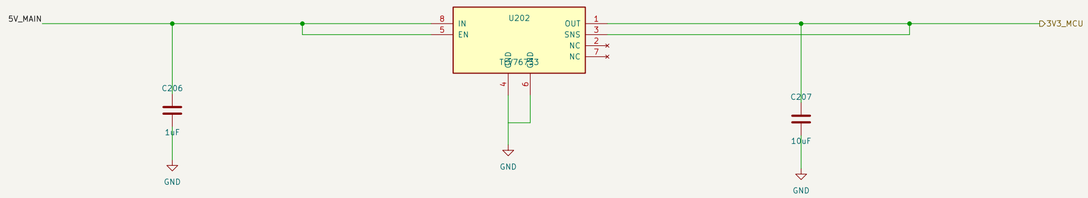
Title: Power input and transient protection

Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 2/11



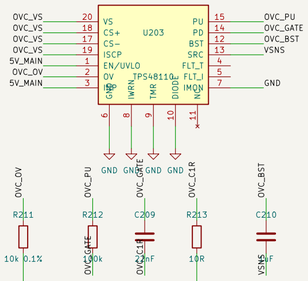


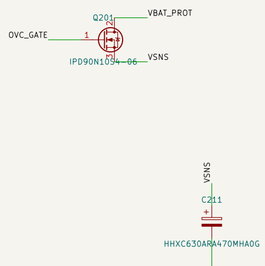
TWO PARTS CHOSEN HERE, both replacing research memo 07 entries that were marked representative rather than chosen:
LM5164 — memo 07 sec.4 names a 60 V TPS54360B-Q1. transient_clamp.cir now carries a PASSING check that reads "60 V-class parts are NO LONGER viable downstream" -- raising the TVS standoff to 43 V, so the part stops conducting during a NORMAL clamped dump, put the pulse 2a clamp at 73.3 V. Every rating check in the suite has been written against the LM5164's own −0.3 V / 100 V VIN limits all along; the BOM was simply never reconciled with it. A 60 V buck here is a part that fails the first pulse 2a event.
TLV76733 — memo 07 sec.6 names TLV1117-33QDCYRQ1, explicitly "class pricing, not fetched". The supervisor chosen on the mcu sheet made 3V3 accuracy load-bearing: outside 3.143–3.459 V the board resets itself. TLV767-Q1 is 1% over load AND temperature, giving 3.267–3.333 V.

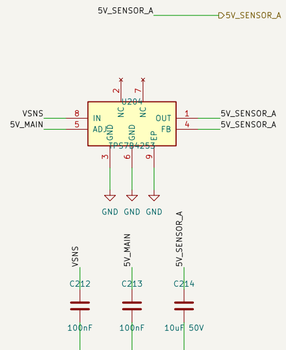


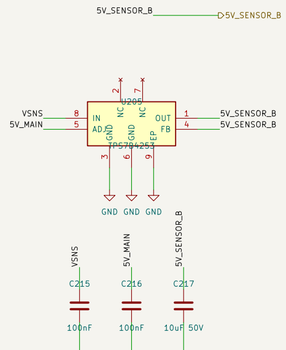
RATIOMETRIC MONITOR: the divider taps 5V_MAIN, and every sensor group's tracker follows 5V_MAIN to +/-4 mV (0.08%) at any load up to 300 mA.
So one tap now corrects all three groups. With the PTCs it could not: each dropped about 150 mV (3%) downstream of the tap.

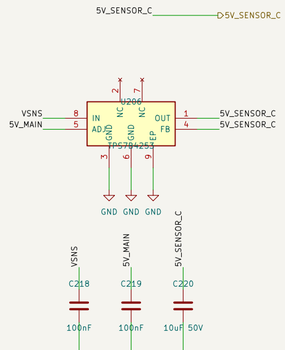
5V_SENSOR -- TPS48110-Q1 overvoltage cut-off (opens above 40.5–42.1 V, closed through the 40 V dump) feeding one TPS7B4253-Q1 tracker per sensor group. sensor_rail.cir.

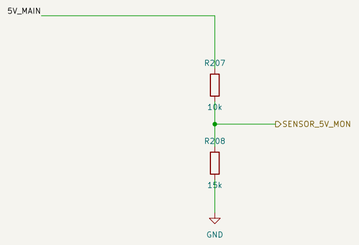












PGOOD_5V is brought out and left unassigned. The LM5164 asserts it open-drain; the pin map has no channel for it. Available, not wired to anything -- shown rather than hidden.

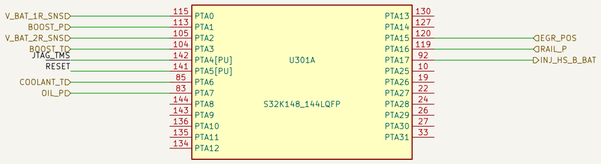
3V3 must stay inside 3.143–3.459 V or the supervisor resets the board
LM5164 replaces memo 07's 60 V TPS54360B-Q1 -- see the note on the page
Buck values: buck_preregulator.cir. 5V_SENSOR: sensor_rail.cir, decisions log 4 and 7
Generated by hw/gen_sheet_rails.py -- do not hand-edit until it is retired

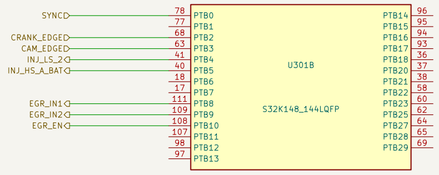
ecu25kva

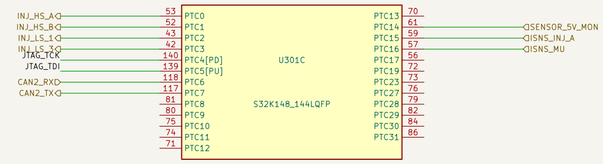
Sheet: /rails/
File: rails.kicad_sch

Title: Rails: buck, 3V3 LDO, 5V sensor distribution

Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 3/11



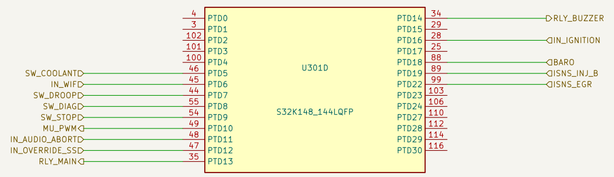


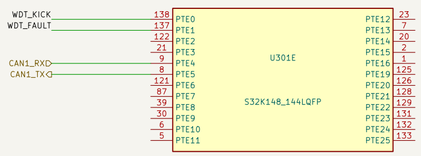


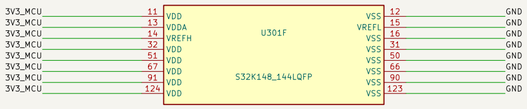
PIN MAP ADDITIONS MADE ON THIS SHEET -- both are gaps in docs/pinmap.md, not changes to it:
  PTE0 WDT_KICK   MCU -> TPS3850 WDI. pinmap.md assigns 37 signals and none of them services the external
                   watchdog, but memo 11's topology requires the MCU to kick it. A window watchdog nobody feeds
                   resets the board on a timer.
  PTE1 WDT_FAULT TPS3850 WDO -> MCU, a live input. D302 makes WDO pull RESET too, so it reads high again
                   once the MCU is out of reset: it does not record which trip it was.
Both are free GPIO-HD pins, PE=0/PS=0 at reset, from the 91 the pin map leaves spare.

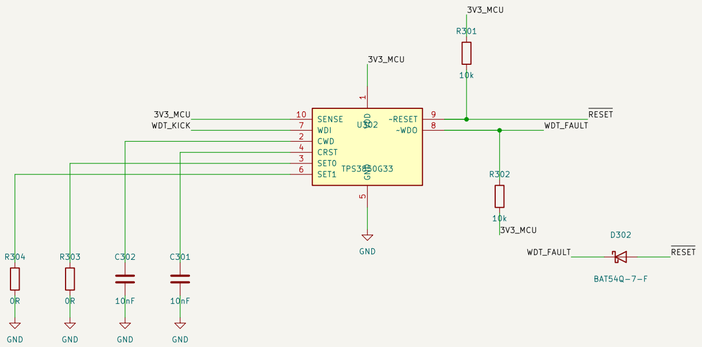
SUPERVISOR CHOICE -- TPS3850G33, from the datasheet's own nomenclature table (G = thresholds at +/-4% of nominal, H = +/-7%) and its +/-0.8% accuracy spec:
  G33 UV 3.168 V nom, 3.143 V worst-low | H33 UV 3.069 V nom, 3.044 V worst-low
The supervisor has to assert BEFORE the MCU's own LVD (3.0 V max) and above the 2.97 V correctness floor. G33 clears LVD by 143 mV worst case; H33 by 44 mV,
close enough that the two could fire in either order. The cost is that the rail must stay inside 3.143-3.459 V -- a +/-2% regulator spans 3.234-3.366 V, leaving ~90 mV each end.
That is this choice's constraint on the rails sheet.

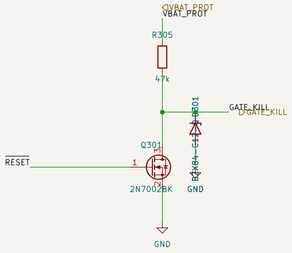
OPEN: SET0/SET1 strap the watchdog window ratio. Firmware now kicks WDI every 10 ms (decisions log 11), so the window must accept 5-20 ms; strap values vs the TPS3850 tables still to check. Link resistors.

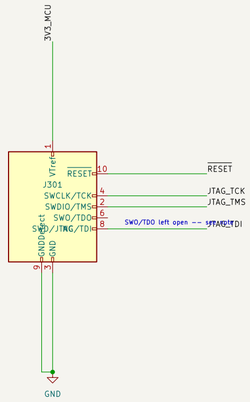


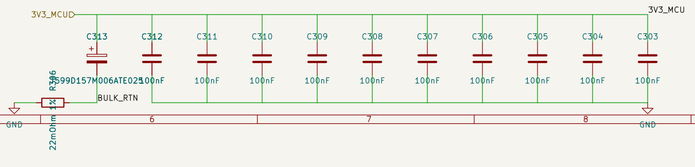












PTE0/PTE1 are added here -- pinmap.md assigns no watchdog pins
Supervisor TPS3850G33: memo 11 sec.2.1 + TI SBVS301B Table 5-1
Pin assignments: docs/pinmap.md. Pin positions: hw/lib/s32k148_layout.json
Generated by hw/gen_sheet_mcu.py -- do not hand-edit until it is retired

ecu25kva

Sheet: /mcu/
File: mcu.kicad_sch

Title: MCU, supervisor and debug

Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 4/11

BOOST RAIL -- 100 V reservoir for the injector peak phase

the converter that feeds this rail is drawn at the foot of the sheet -- TPS40210-Q1, VBAT_PROT to BOOST_100V

bank A (cyl 1+3)

bank B (cyl 2)

GATE_KILL reaches the four high sides at the drivers' HIN inputs, not at the gates -- a ground-referenced kill FET cannot short a gate floating 100 V up, but HIN is ground-referenced. See the gate-driver block on the right.

GATE DRIVERS -- AUIRS2181S x4. Each pairs one high side with one low side; the pairing is by package, not by circuit.

A GROUND-REFERENCED KILL CLAMP CANNOT SERVICE A HIGH-SIDE GATE. and spec sec.4's fail-safe list names pins 03/05 alongside the four low-side gates as though it could. The three low sides here get exactly the network supervisor.clr sized and docs/pinmap.md sec.2 confirmed pin by pin: a 470 ohm pulldown to ground, plus a small-signal FET that shorts the gate to ground when GATE_KILL asserts, with no MCU involvement -- which is what makes it work against a hung MCU and not only a reset one.

The FOUR high sides cannot use it -- two per bank since the 21 Sep correction. The source of a high-side N-FET swings to the boost rail, so a pulldown to GROUND would hold Vgs at minus the source voltage and fight the driver on every edge, and a kill FET referenced to ground cannot short a gate that is floating 100 V up. What is drawn instead is the 470 ohm across GATE and SOURCE -- the same Miller-coupling divider supervisor.clr sized, just referenced to the terminal it has to be referenced to -- and the kill path moves into the driver: GATE_KILL must reach the high-side driver's LOGIC INPUT. AUIRS2181S has no shutdown pin, so the same kill FET sits on HIN behind a 1k from the MCU -- the one ground-referenced point in that path. That is drawn in the gate-driver block, and it is the sixth finding of the shape this project keeps turning up: a document and an executable file agreeing with each other and both wrong about a case neither modelled.

GATE DRIVERS: AUIRS2181S x4, chosen 21 Sep 2026. Its first listed application is 'Piezo / common rail injection', and three properties decided it, each against a part that failed on it. NO CROSS-CONDUCTION INTERLOCK: in a half bridge HO and LO together is shoot-through, but here the high and low switches are in SERIES with the coil and must BOTH be on to fire it -- UCC27282-Q1 and UCC27712-Q1 have interlock and could not fire an injector at all. 600 V OFFSET, against UCC27211A-Q1's 120 V HB absolute maximum, which the bank node plus 12 V clears by 4 V. VS OPERATIONAL TO -5 V, against UCC27211A-Q1's -1 V HS DC minimum, which D_fw violates on every hold off-time at 1.0-1.3 V. Each package pairs one high side with one low side; the pairing is by package, not by circuit, and UB2's low side is spare.

12V_GATE: the board had no rail between 5 V and 40 V, and the drivers need 10-20 V. It is made from the BOOST rail, because a regulator from the battery cannot make 12 V from a 6-9 V cranking battery -- and a gate rail that collapses during cranking means no injection exactly when the engine needs it. Not the LM5164 already on the board: its VIN absolute maximum is 100 V and this rail sits at ~102 V. A discrete follower with a current limit, for a ~3 mA load. If the boost stops, every driver drops into UVLO and every gate turns off -- the failure runs the safe way.

STILL NOT CHOSEN: the two current-sense amplifiers (ISNS_INJ_A/B_SENSE -> PTC15 / PTD19). Ground-referenced -- that is why the shunts went low-side -- gain about 33, 100 kHz or better; and a Kelvin tap at the shunt pad.

THE BOOST CONVERTER IS NOW DRAWN, AND THE PART WAS CHOSEN ON ONE NUMBER: tOFF(min) = 200 ns max, TI SLVSB61F Table 6.6. A boost needs 94.1% duty at the 6 V cranking corner, and at 150 kHz that off-time floor allows 97.0%. Most wide-input boost controllers in this class switch at 2.2 MHz, where the same 200 ns caps duty at 56% -- below even the 86.6% the rail needs at a nominal 13.5 V battery. Switching SLOWER is the requirement here, which runs the opposite way to the usual 'higher frequency, smaller magnetics'.

THE RATING THAT DOES NOT FIT, and it is the third time this board has hit it: VDD absolute maximum is 52 V (Table 6.1) and VBAT_PROT reaches 73.3 V for about 50 us on ISO 7637-2 pulse 2a -- over by 1.41x. Same shape as DRVB873-Q1's 40 V VM, which took that part off the EGR bridge. It is solvable here for a reason that did not apply there: VDD draws 2.5 mA plus gate charge, not a motor's current, so 220 ohm and a 36 V zener hold the pin at 46.6 V worst case through the pulse -- the zener's tolerance, tempco and dynamic resistance included. (47 ohm and an ideal 43 V were drawn first; the real part gave 61.9 V.) On the 400 ms load dump the clamp passes 33 mA at most. The POWER stage is untouched: the inductor and FET see the rail directly and are 150 V class for it. All of this is checked in run_sim.py under boost_converter -- not with a switching model, which that deliberately does not have, but as executable arithmetic on the datasheet's own ratings, the same treatment transient_clamp.clr gives the 73.3 V rail.

BOOST CONVERTER -- TPS40210-Q1, 6-40 V in, 100 V out at 50 mA average. The reservoir above delivers the pulse; this delivers the trickle.

VDD FOLLOWER (BOOST-SLOPE) -- the TPS40210-Q1 runs from its own output once started: VDD 34.3-45.0 V. The battery path only starts it.

12V_GATE -- gate-drive supply for every driver on the board, from the boost rail so it survives cranking

THE COMPENSATION VALUES ARE A STARTING POINT AND THE SHEET SAYS SO. Rcomp 10k and Ccomp 6.8 nF put the error amplifier's zero at 2.3 kHz, below the right-half-plane zero a boost converter puts at (1-D)^2 * Rload / (2*pi*L) = 17.3 kHz at 13.5 V in -- which is the constraint that actually bounds crossover, and it MOVES with duty cycle, so the cranking corner is the hard one. Everything else on this sheet traces to a datasheet number or a simulation result. These two do not, and cannot: a current-mode boost's loop depends on the inductor's real DCR, the capacitor's real ESR and the FET's real switching behaviour. They are bench measurements. What is defensible now is the TOPOLOGY and the starting values; the loop gets measured.

NOT DRAWN, and deliberately: no output capacitor appears here, because the 41 uF reservoir at the top of this sheet IS the output capacitor. Its 22k bleed is the only thing that can take charge OFF this rail -- a boost converter cannot pull its own output down, which is the same one-sided regulation boost_converter.clr's arrangement C measures.

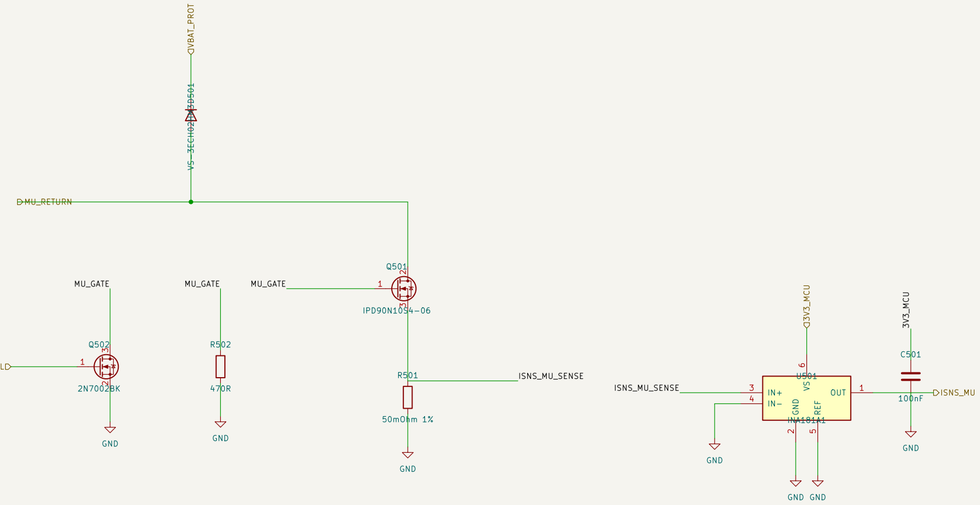
CURRENT SENSE -- INA181A1-Q1 per bank, gain 20

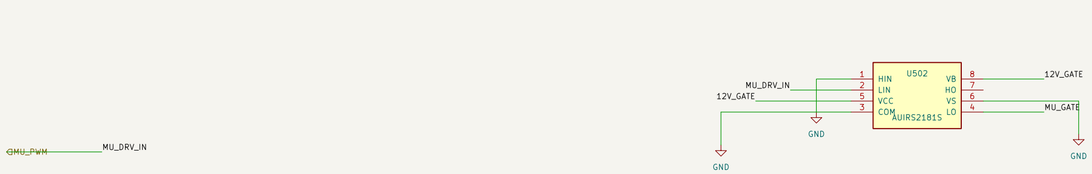
Boost controller TPS40210-Q1, chosen on its 200 ns off-time; 150 kHz reaches 94% duty at cranking
Hold current arrives through a SECOND high-side switch per bank -- corrected 21 Sep 2026
boost_converter.clr / injector_boost.clr / injector_turnoff.clr
Generated by hw/gen_sheet_injector.py -- do not hand-edit until it is retired
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Sheet: /injector/
File: injector.kicad_sch

Title: Injector drive: boost converter, boost rail, high sides, low sides

Size: A1	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 5/11

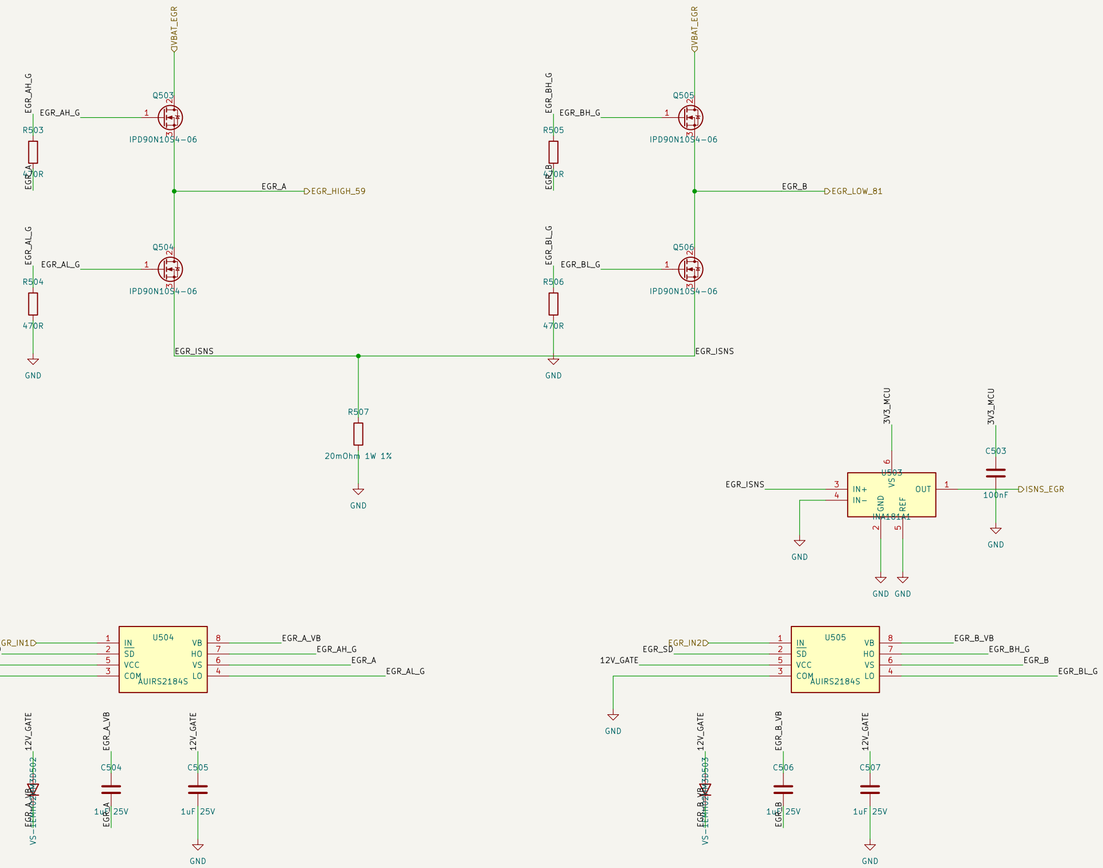




MU_GATE IS DRIVEN BY AN AUIRS2181S, low side only -- the same part as the four injector drivers, so the board carries one gate-driver part number. What it had to be, from supervisor.cir claim 2: output impedance <= 25 ohm, because the 470 ohm pulldown is the bottom half of a divider with it. The datasheet's 1.4/1.8 A minimum short-circuit current at 15 V puts it at roughly 8-11 ohm, so the gate reaches 11.7 V rather than being divided down. VCC is 12V_GATE, made on the injector sheet from the boost rail so that it survives cranking.

ISNS_MU_SENSE goes to a current-sense amplifier, also not chosen. 50 mOhm at the 0.675 A setpoint is 33.75 mV, measured ACROSS the shunt -- the amplifier's low reference is the shunt's own ground pad, which is electrically GND and so is not a separate net. It IS a layout constraint: a Kelvin tap at the pad, not a via into the ground pour, because 33.75 mV sits on a return carrying up to 1.35 A and a few milliohms of pour between pad and reference is the whole signal.
Output goes to PTC16 (ADC0_SE14). The loop is closed in firmware: PTD10 carries the PWM, PTC16 reads the current back. That is arrangement the pin map already assumes, and it is why metering_unit_pwm.cir models the regulator as hysteretic rather than as a fixed duty.

EGR BRIDGE -- discrete, two AUIRS2184S. Coast (both SD-bar low) is the reset state and the kill state.



WHY DISCRETE, AND WHY THE 2184. DRV8873-Q1 was the candidate and had exactly the decoded interface EGR-DRIVER asks for -- and a VM absolute maximum of 40 V against a rail that reaches 73.3 V. Integrated automotive bridges cap out at 40-45 V as a class. So four 100 V FETs, and the decode comes from the driver's STRUCTURE: AUIRS21845 has one input per leg, so each leg's high and low are exclusive by construction. egr_hbridge.cir's 270 A state -- both inputs high, all four on -- becomes a high-side BRAKE. Same family as the injector drivers, chosen the opposite way for the opposite reason: the injector's switches are in series with a coil and must BOTH conduct.

IN1 IN2 with EGR_EN high: 1 0 forward 0 1 reverse 1 1 high-side brake 0 0 low-side brake. EGR_EN low: COAST, whatever IN1/IN2 are.

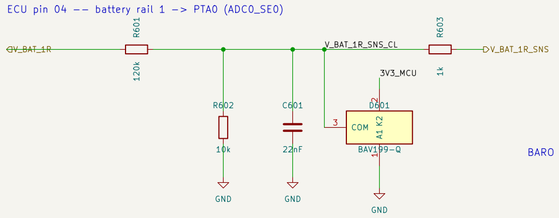
Nothing battery-connected on this board may be rated below 80 V
EGR half NOT drawn -- blocked on a battery-rail rating, see the note
Metering unit: metering_unit_pwm.cir + bom_requirements MU-ISENSE, GATE-PULLDOWN
Generated by hw/gen_sheet_metering_egr.py -- do not hand-edit until it is retired
ecu25kva

Sheet: /metering_egr/
File: metering_egr.kicad_sch

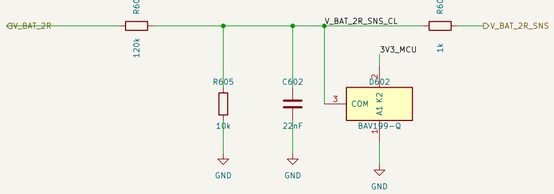
Title: Fuel metering unit and EGR actuator

Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 6/11

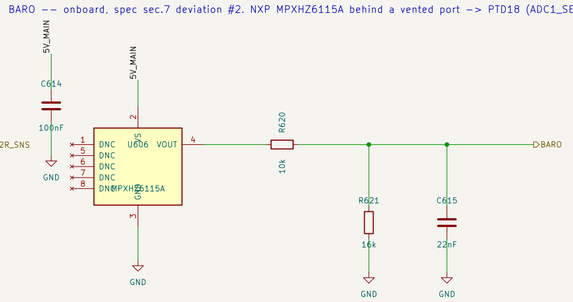
ECU pin 04 -- battery rail 1 -> PTA0 (ADC0_SE0)



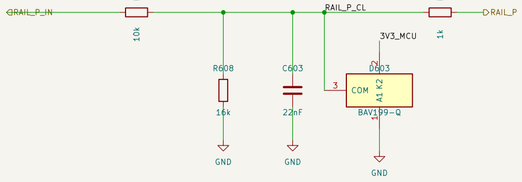
ECU pin 06 -- battery rail 2 -> PTA2 (ADC1_SE0)



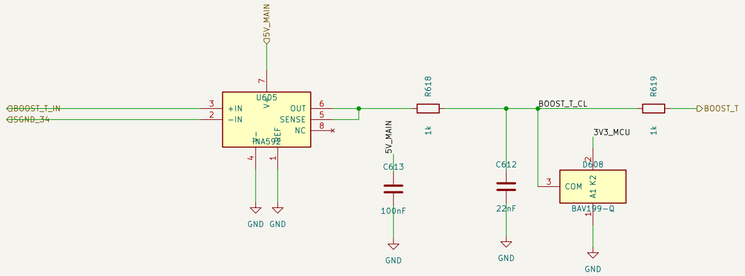
BARO -- onboard, spec sec.7 deviation #2. NXP MPXHZ6115A behind a vented port -> PTD18 (ADC1_SE16)



ECU pin 35 -- rail pressure, DEDICATED return pin 08 -> PTA16 (ADC1_SE13)



ECU pin 79 -- boost temperature -> PTA3. U2 OPEN: sensor type unmeasured



5V_SENSOR excitation -- three PTC-protected groups from the rails sheet, out to five connector pins

WHY FIVE CHANNELS ARE DIFFERENTIAL AND ONE IS NOT. The split follows the harness, not the sensor. Boost pressure and coolant temperature SHARE connector pin 34 as their return: EGR position and oil pressure share pin 36. Rail pressure has pin 08 to itself. A shared return carries both sensors' current, so its own resistance appears as an offset on both readings -- sensor_differential.cir puts 20 mA down a 1 ohm corroded return and measures a single-ended reading wrong by the whole offset while the differential one rejects it, a ratio of 90x. A dedicated return does not have that problem, so rail pressure gets the simpler front end.

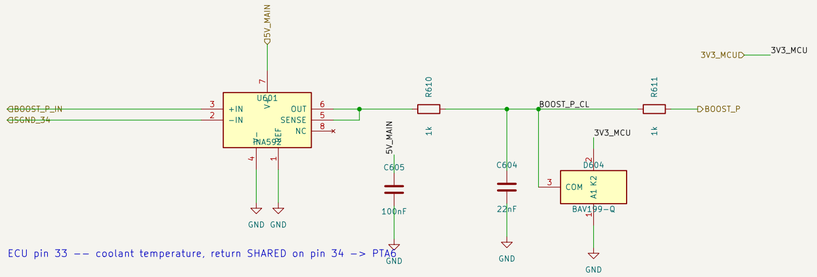
The block states the consequence as a harness requirement, which is the useful form: single-ended is only acceptable if contact resistance is maintained below a figure nobody is going to maintain over a genset's life in an Indian monsoon.

DIFFERENCE AMPLIFIERS: TI INA592 at its native G = 1/2, chosen 22 Sep 2026. The requirement was >=60 dB CMRR -- the floor sensor_differential.cir tried -- and 0.1% discrete resistors give about 48 dB, so the network had to be matched on-die: 88 dB minimum. G = 1/2 puts a 0.5-4.5 V sensor at 0.25-2.25 V: output reaches within 220 mV of each rail. A 1k before each ADC tail limits the pin zener to 1.5 mA if an amplifier rails on a harness fault. NOT AEC-Q100: no qualified G = 1/2 difference amplifier was found, and INA2132's 'G = 1/2' is a single-ended attenuator, not a difference stage.

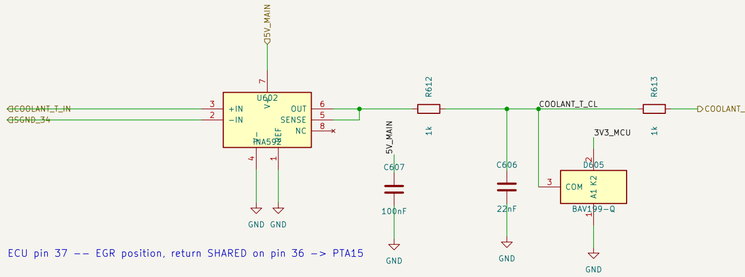
BOOST TEMPERATURE (PTA3, connector pin 79) is drawn differential like its neighbours, but U2 is still open: whether that element is a ratiometric sensor or a bare NTC is unmeasured. ntc_frontend.cir already chose between the two NTC options if it turns out to be one -- a 2.2k pull-up to 5 V, not a 100 uA current source, because the source needs 5.9 V of compliance at -40 C and the rail is 5 V. That is a populate-time change to this channel, not a redesign.

BARO (PTD18) is drawn: NXP MPXHZ6115A, spec sec.7 deviation #2, on-PCB behind a vented port. AEC-Q100 is not stated in its retrieved datasheet; Infineon KP236 is the qualified alternative whose datasheet did not resolve -- a footprint change if chosen.

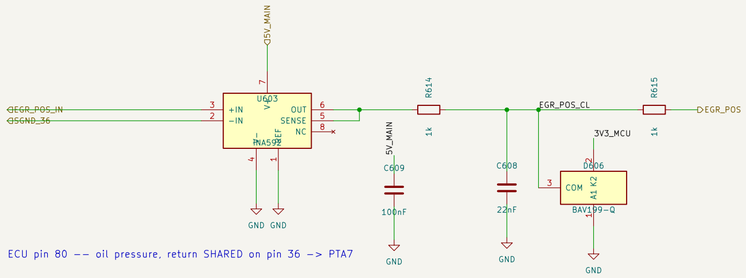
ECU pin 41 -- boost pressure, return SHARED on pin 34 -> PTA1



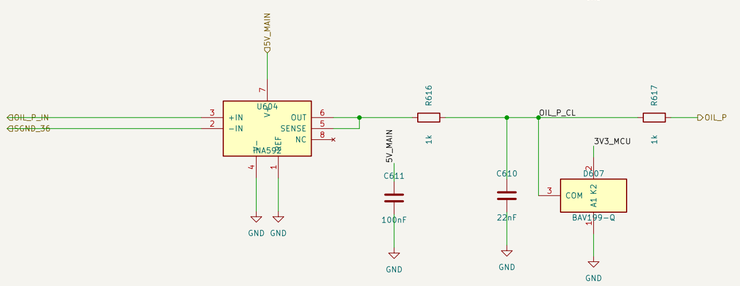
ECU pin 33 -- coolant temperature, return SHARED on pin 34 -> PTA6



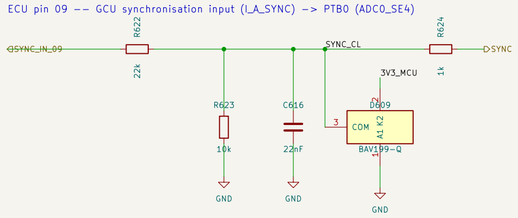
ECU pin 37 -- EGR position, return SHARED on pin 36 -> PTA15



ECU pin 80 -- oil pressure, return SHARED on pin 36 -> PTA7



ECU pin 09 -- GCU synchronisation input (I_A_SYNC) -> PTB0 (ADC0_SE4)



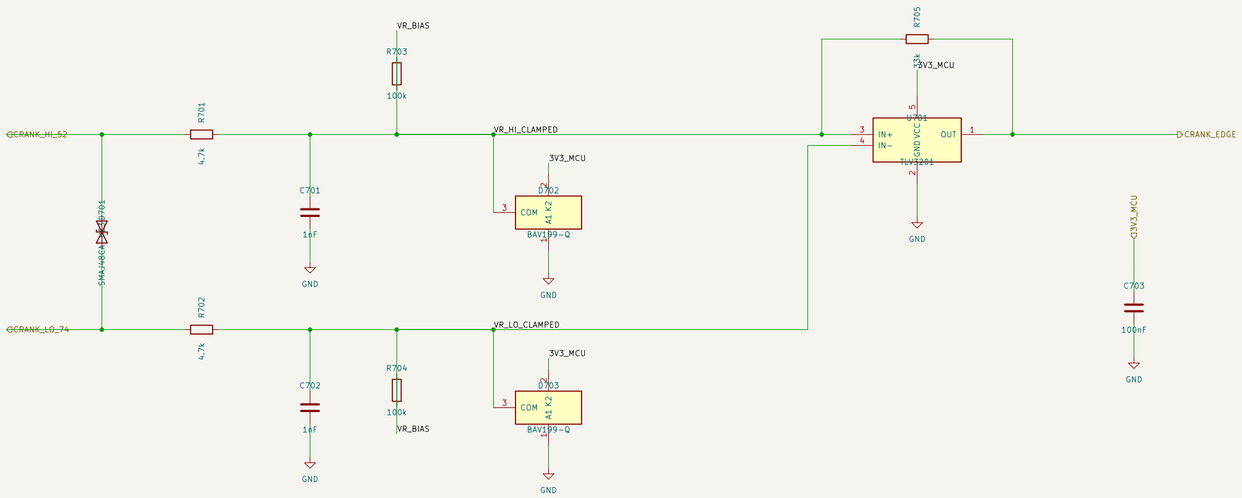
Difference amplifiers are INA592 at G = 1/2: 88 dB CMRR min against a 60 dB floor
Differential where the sensor return is SHARED; single-ended where it is dedicated
battery_sense.cir / sensor_ratiometric.cir / sensor_differential.cir / ntc_frontend.cir
Generated by hw/gen_sheet_sensors_analog.py -- do not hand-edit until it is retired
ecu25kva

Sheet: /sensors_analog/
File: sensors_analog.kicad_sch

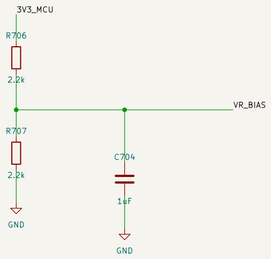
Title: Analog sensor front-ends

Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 7/11

CRANK -- variable reluctance, ECU pins 52 / 74, shield 30 -> zero-cross comparator -> PTB2 (FTM1_CH0)



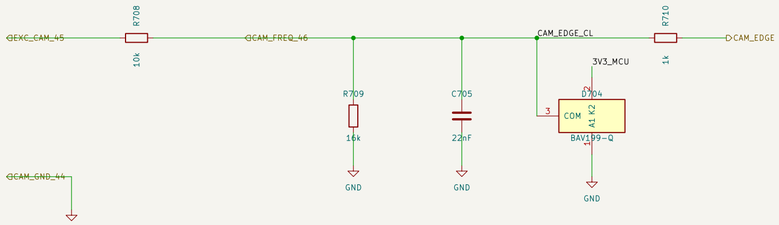
VR_BIAS -- mid-rail reference for the floating VR coil



ECU pin 30 -- crank sensor shield, terminated at THIS end only. Grounding both ends makes the shield a second conductor between two grounds that differ by the injector return current, and that current then flows down the shield around the pair it is there to protect.



CAM -- digital Hall, ECU pins 45 / 46 / 44 -> PTB3 (FTM1_CH1). CORRECTS the front-end spec sec.4 specifies literally



CLAMP LEAKAGE IS THE BINDING SPEC, WHICH INVERTS THE RULE THIS PROJECT USED ON THE POWER PATH. Every earlier clamp on this board was chosen for forward drop -- the negative-clamp Schottky's 0.547 V at 15 A decomposes into junction and bulk terms and that decomposition is what ruled out paralleling junction devices. Here the forward drop does not matter at all: these diodes conduct only on a fault, into a 4.7k series resistor that is already doing the limiting. What matters is what they do when they are OFF.

A reverse-biased diode's leakage flows out of the 4.489k Thevenin at the input node and appears as an offset voltage there. The hysteresis window is +/-198 mV, so:

BAT54S-class Schottky ~2 uA at 25 C, and Schottky leakage rises roughly 100x to 125 C -> ~200 uA -> 0.90 V of offset.
That is 4.5x the whole hysteresis window. The comparator latches and the engine has no speed signal at temperature.
BAV199-class low-leakage silicon 5 nA max at 25 C, under 1 uA at 125 C -> under 4.5 mV of offset, 2% of the window.

The two clamp diodes are biased symmetrically (both reverse-biased by 1.65 V from VR_BIAS) so their leakages partly cancel, and both legs are matched so what survives is partly common-mode -- but a design whose speed input depends on cancellation between two diodes' hot leakage is not one to defend. The requirement is stated instead: reverse leakage at or below 1 uA at 125 C. vr_conditioner.cir does NOT model this -- its comparator is an ideal switch with no input network at all, so no clamp, no leakage and no offset exist in it. This is the sheet finding something the block could not.

WHY THE CAM FRONT-END ON THIS SHEET IS NOT THE ONE THE SPEC ASKS FOR. Spec sec.4 and docs/pinmap.md sec.1.3 both give pin 46's front end as "Pull-up to 5V_SENSOR -> timer capture": the Hall output pulled to 5 V and taken straight into PTB3. NXP S32K1xx Data Sheet Rev. 15, Table 17, puts Vih max at VDD + 0.3 V, about 3.6 V at VDD = 3.3 V, so a full 5 V swing is not a guaranteed valid logic level. Table 1 does allow -0.8 to +5.8 V on any I/O pin, but only while respecting the +/-3 mA injection limit, and the datasheet's own words are that "functional operation at the maximum values is not guaranteed" -- a stress margin, not a design basis for a pin that would sit at 5 V once per cam revolution for the machine's running life. The 16k to ground turns it into the same 10k/16k divider the ratiometric sensor channels already use. It costs one resistor.

This was found by writing cam_frontend.cir, which pinmap sec.4 asked for by name before this sheet was drawn: no block had ever simulated a cam front-end -- vr_conditioner.cir covers the crank and there was no cam equivalent -- so a 5 V swing on a 3.3 V pin had never been checked. Fourth finding of the same shape in this project: a document and an executable file disagreeing, with nothing comparing them.

PARTS: the comparator (now TLV3201-Q1, the clamp comparator on power_input) and the cam Hall sensor itself, which is U8 on the unknowns register and needs the machine. The comparator's requirement is unusually easy, and deliberately so: biasing the floating coil to VR_BIAS means the inputs never leave mid-rail, so no part is needed whose common-mode range includes ground -- the usual hard constraint on a zero-cross detector. Single 3.3 V supply, push-pull rail-to-rail output, propagation delay at or below 1 us (0.009 crank degrees at rated speed on a 60-tooth wheel). Input offset voltage is a non-issue against a 198 mV window.

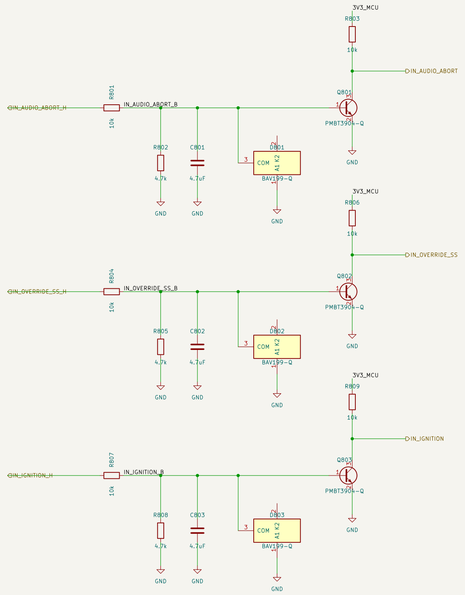
Clamp diodes specified by REVERSE LEAKAGE, not forward drop -- see the sheet note
Zero-cross with +/-198 mV hysteresis: a fixed threshold cannot see a 2 V cranking signa
vr_conditioner.cir / cam_frontend.cir
Generated by hw/gen_sheet_speed_inputs.py -- do not hand-edit until it is retired

ecu25kva

Sheet: /speed_inputs/
File: speed_inputs.kicad_sch

Title: Crank VR conditioner and cam front-end

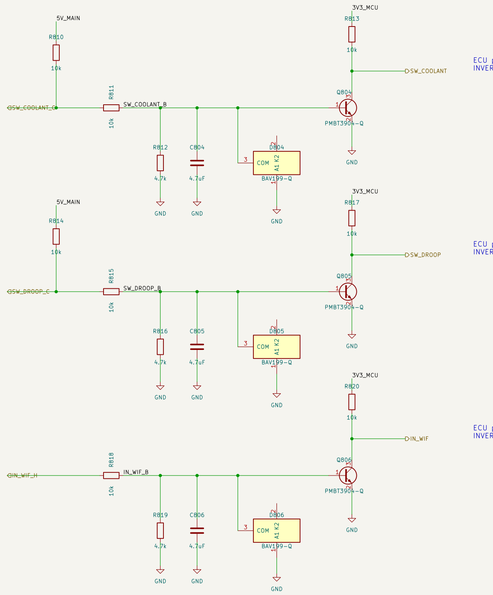
Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 8/11



ECU pin 20 -- audio abort, active high -> PTD11
INVERTED: pad LOW = input active

ECU pin 24 -- override start/stop, active high -> PTD12
INVERTED: pad LOW = input active

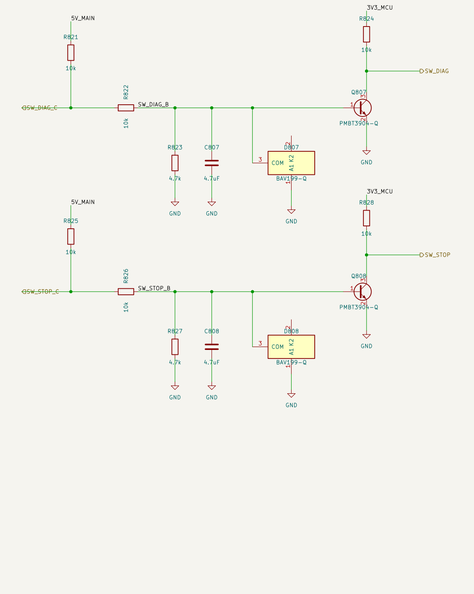
ECU pin 71 -- ignition, active high -> PTD16
INVERTED: pad LOW = input active



ECU pin 43 -- coolant switch, contact to ground -> PTD5
INVERTED: pad LOW = input active

ECU pin 23 -- droop switch, contact to ground -> PTD7
INVERTED: pad LOW = input active

ECU pin 21 -- water in fuel, active high -> PTD6
INVERTED: pad LOW = input active



ECU pin 14 -- diagnosis request, contact to ground -> PTD8
INVERTED: pad LOW = input active

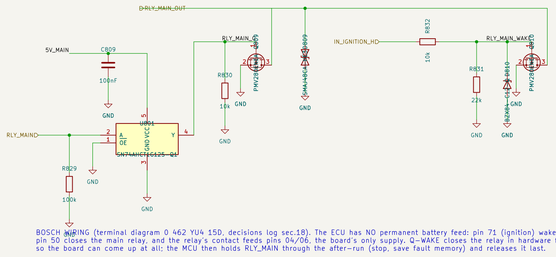
ECU pin 25 -- stop / emergency switch, contact to ground -> PTD9
INVERTED: pad LOW = input active

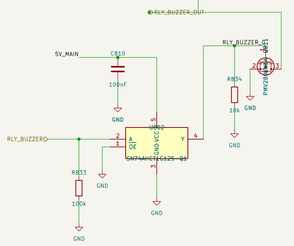
ECU pin 50 -- main relay, low side <- PTD13, woken by pin 71

ECU pin 69 -- buzzer relay, low side <- PTD14

3V3_MCU 3V3_MCU

5V_MAIN 5V_MAIN





BOSCH WIRING (terminal diagram 0 462 YU4 15D, decisions log sec.18). The ECU has NO permanent battery feed: pin 71 (ignition) wakes it,
pin 50 closes the main relay, and the relay's contact feeds pins 04/06, the board's only supply. Q-WAKE closes the relay in hardware from pin 71,
so the board can come up at all: the MCU then holds RLY_MAIN through the after-run (stop, save fault memory) and releases it last.

PIN 50 HAS A TVS, NOT A FLYBACK DIODE. Its coil sits on Term. 30, always live. A diode from the drain to VBAT_PROT powered the board THROUGH THE COIL
with the relay open: 22.8 mA in the coil (2.7x its release current) and the board at 9.1 V (main_relay.cir part B). The relay could never drop out.
Bosch marks pin 50 'not equipped with freewheeling diode'. The buzzer relay keeps its diode: its coil is on the main-relay line and dies with VBAT_PROT.

NO TRIP LOOP. It had no Bosch pin. The overspeed trip module (external) opens the main relay's coil feed, which drops the board with everything else.
Pin 21 is the WATER-IN-FUEL input, active high from the main-relay line through the sensor's switch, open (reads inactive) with no water.

THE EIGHT SWITCH INPUTS ARE NPN BUFFERS, NOT CLAMPS (22 Sep 2026, LOWV-CLAMP). The 47k/68k/3.0 V-zener front end needed a clamp that conducts CONTINUOUSLY over 6-40 V, and none
fits the pad's ratings: a 3.0 V zener leaks at uA (soft knee), a rail clamp parks the pad at 3.9 V, a shunt reference draws 35 uA of bias the cranking-dip input cannot supply.
Here the pad sees only its own 10k pull-up or a saturated PMBT3904-Q. Checked at the WORST corner, not typical: hFE 30 at -40 C still pulls the pad to 0.03 V from a 6 V input;
a typical part at 125 C does not turn on below 1.3 V. 10k input resistors give 1.1 mA of contact wetting current at 12 V.

LOGIC IS INVERTED -- pad LOW = input active (contact-to-ground channels; contact OPEN). FAILURE MODES, for firmware plausibility checks: transistor open or pull-up shorted
-> pad stuck HIGH (reads inactive); C-E short -> pad stuck LOW (reads active). One polarity cannot make every channel fail safe; the safe state per channel belongs in the
firmware spec. The trip loop keeps its clamped ANALOG front end: its healthy band is the clamp level (LOWV-CLAMP residual).

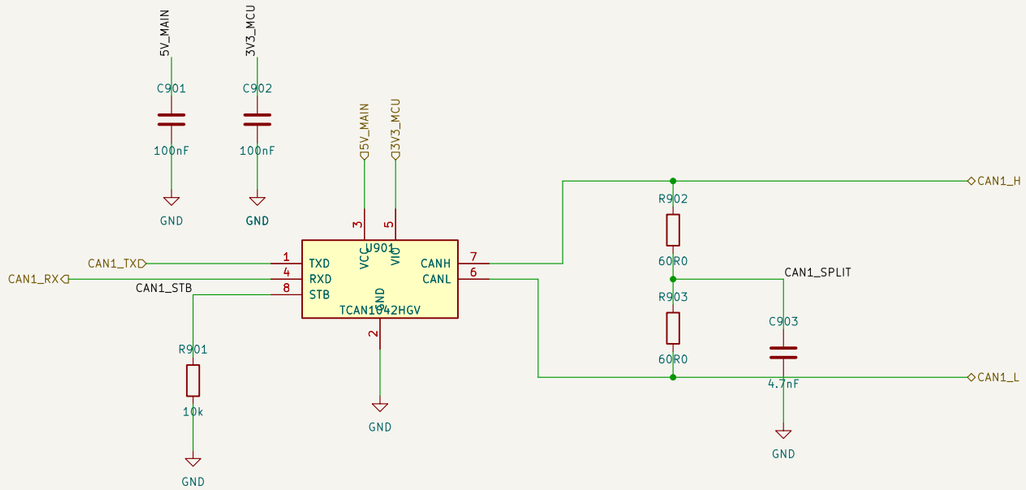
Relay FETs are 100 V class -- VBAT_PROT reaches 73.3 V on pulse 2a
Bosch pinout (refs/bosch-pinout-m81cs162.md): pin 71 wakes, pin 50 holds, pin 21 is water-in-fuel
Inputs: NPN buffers (discrete_input.cir). Inverted. Outputs: relay_driver.cir, main_relay.cir
Generated by hw/gen_sheet_discrete_io.py -- do not hand-edit until it is retired

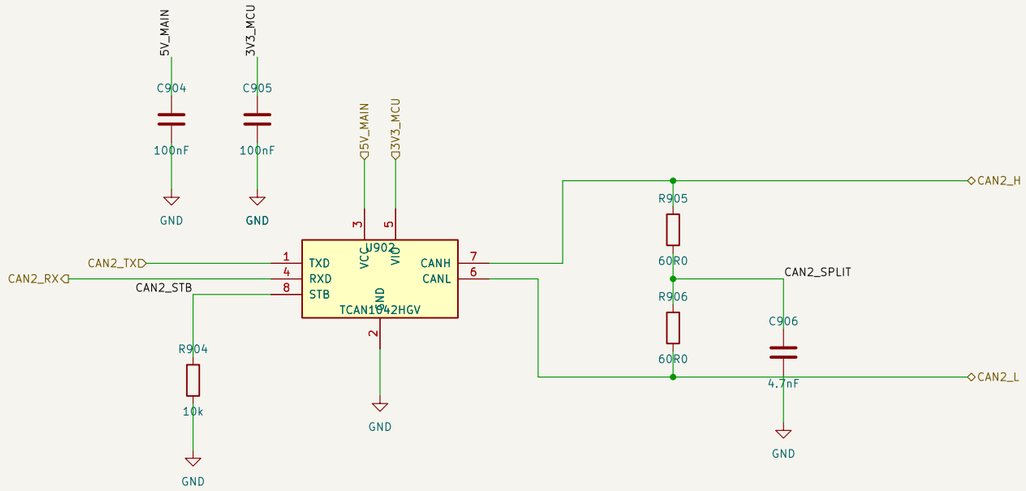
ecu25kva

Sheet: /discrete_io/
File: discrete_io.kicad_sch

Title: Discrete I/O: switch inputs, relay drivers, ignition wake

Size: A1	Date:	Rev:
KiCad E.D.A. 10.0.6		id: 9/11





TERMINATION IS A FIT OPTION ON BOTH CHANNELS, and that is a real open question rather than caution.
J1939 terminates the two ENDS of a backbone, not every node. Channel 1 is almost certainly a two-node link to the fitted DSE4522 -- which takes engine speed from this ECU over CAN -- so both ends terminate and fitting it here is right. Channel 2's harness topology is not established. Fitting 120 ohm at a mid-bus node over-terminates the backbone and costs signal amplitude at every other node on it, so the resistors are drawn but flagged, and the decision is the harness's, not this board's.

WHY SPLIT AND NOT ONE RESISTOR -- can_termination.cir. Both topologies are 120 ohm differentially and measure IDENTICAL there, which is why a schematic label cannot tell them apart. They differ only in common mode: a single 120 ohm connects CANH to CANL and nothing else, so common-mode noise -- both wires swinging together against chassis, which is what an alternator and contactors produce -- sees an open circuit and stays on the bus. Measured at 10 MHz: 30 ohm for the split network against more than 500 kohm for the single resistor, a factor of 16562. At the 250 kbit/s bit rate the midpoint is still 138 ohm, so it shunts the noise without loading the signal.

Termination is a FIT OPTION on both channels -- see the note
Thin-film 0.1% / 50 ppm -- thick-film misses J1939's 120 ohm at temperature
Termination: can_termination.cir + bom_requirements CAN-TERM
Generated by hw/gen_sheet_can.py -- do not hand-edit until it is retired
ecu25kva

Sheet: /can/
File: can.kicad_sch

Title: CAN: two J1939 channels, split-terminated

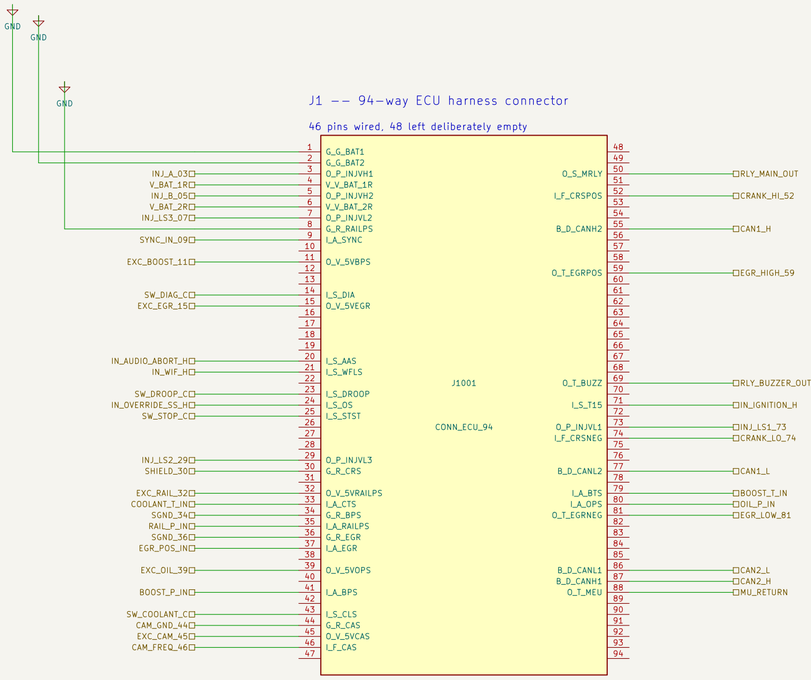
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 10/11

POWER, AS BOSCH WIRES IT.

There is no permanent battery feed. The main relay's coil sits between Term. 30 and pin 50; its contact feeds pins 04 and 06 through two 10 A fuses, and those are the board's only supply. Pin 71 (ignition) closes the relay in hardware through the wake FET on discrete_io, so the board can come up at all. The MCU then holds pin 50 through the after-run and releases it last.

Pin 04 feeds the electronics (power_input's 5 A fuse, EMI filter, TVS, reverse FET, buck) AND the injector stage: pin 06 the EGR bridge. 01 and 02 are the power ground (G_G_BAT1/2): the 18 A injector return lands there.

Pin 21 is NOT battery. It is the water-in-fuel switch, active high from the main-relay line, open with no water.



PINS ADDED FROM THE BOSCH DIAGRAM

09 I_A_SYNC analog synchronisation input from the GCU -> sensors_analog divider -> PTB0
14 I_S_DIA diagnosis request, GCU contact to ground
21 I_S_WFLS water in fuel, active high
25 I_S_STST stop / emergency switch, contact to ground
01 G_G_BAT1, 02 G_G_BAT2 power ground

CAN: Bosch's CAN 1 (application line, 1000 kbit/s, flashing and UDS) is 87/86, net CAN2_*; its CAN 2 (PT line, 250 kbit/s, J1939 and OBD) is 55/77, net CAN1_*. Names kept, mapping noted.

WHY PINS 34 AND 36 ARE SIGNALS AND PIN 08 IS GROUND

34 and 36 are SHARED sensor returns -- 34 for boost pressure and coolant temperature, 36 for EGR position and oil pressure. A shared return carries both sensors' current, so its own resistance is an offset on both readings, which is why sensors_analog carries them into difference amplifiers rather than to ground. 08 is rail pressure's DEDICATED return: nothing to reject, so it is simply ground here.

THE PART IS IDENTIFIED; THE LAND PATTERN IS NOT. refs/field-evidence-2026-09-18.md sec.2b closed U1 off a photograph of the housing angled to the light: BOSCH and the armature-in-circle logo moulded on the side body, 1 928 405 194 under the BOSCH text, 1 928 405 192 on the lower body strip, CODE C beside it, >PA66-GF50< on the locking lever. Read off the physical part, which beats a catalogue match because it cannot be a mis-identification. CODE C IS LOAD-BEARING: Bosch supplies these housings mechanically coded so differently-coded plugs cannot mate, so any replacement or adapter must match it or it will not engage -- the kind of detail that is invisible until a part arrives and will not fit. The PIN LIST is Bosch's terminal diagram (8 Nov 2026). Still missing: the board-side footprint. A search for both numbers returned no catalogue hit, so the land pattern has to come through a Bosch distributor quoting them, or off the old ECU's header (brief Group 6). Nothing is assigned here, because a mating land pattern is not a thing to infer from a photograph.

Bosch 1 928 405 192 / 194 code C ON THE BOARD, no adapter (owner, 22 Sep) -- land pattern still
Power: pin 71 wakes, pin 50 closes the main relay, 04/06 feed the board, 01/02 are ground
Pinout: Bosch terminal diagram 0 462 YU4 15D via refs/bosch-pinout-md1cs162.md (46 pins)
Generated by hw/gen_sheet_connector.py -- do not hand-edit until it is retired

ecu25kva

Sheet: /connector/
File: connector.kicad_sch

Title: 94-way ECU connector and harness mapping

Size: A2	Date:	**Rev:**
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